

Exploratory Data Analysis Report

Customer Churn Prediction & Retention Intelligence System

Prepared for: Sarah Ahmed, Online Fashion Store

Prepared by: Muhammad Huzaifa

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50,000 Original Records	49,584 Unique Customers	416 Duplicate Records	310 Invalid Future Dates
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Executive Summary

This report summarizes the exploratory data analysis conducted for Sarah Ahmed’s online fashion store. The goal of the analysis was to understand customer behavior, identify data quality issues, and determine which variables are most useful for predicting customer churn.

Core Finding

Customer churn risk is driven more by behavioral and engagement signals than by demographics. Purchase frequency, website activity, email engagement, and support ticket volume showed the clearest relationship with churn-related behavior.

Report Contents

- Project and dataset overview
- Data quality findings and cleaning decisions
- Feature-level exploratory analysis
- Key business insights and modeling recommendations
- Final EDA conclusion and next steps

1. Project Overview

Sarah Ahmed’s online fashion store wants to reduce customer loss and improve the return on marketing campaigns. Instead of sending discounts to all customers, the business requires a data-driven system to identify customers who are likely to stop purchasing and prioritize them for retention actions.

The EDA phase focused on understanding the dataset, detecting issues that could affect model reliability, and identifying behavior patterns associated with churn risk.

Business Objectives

- Identify customers who are likely to stop purchasing.
- Improve marketing ROI by targeting high-risk customers instead of all customers.
- Increase repeat purchases through proactive retention actions.
- Generate a reliable customer-level prediction report for monthly business use.

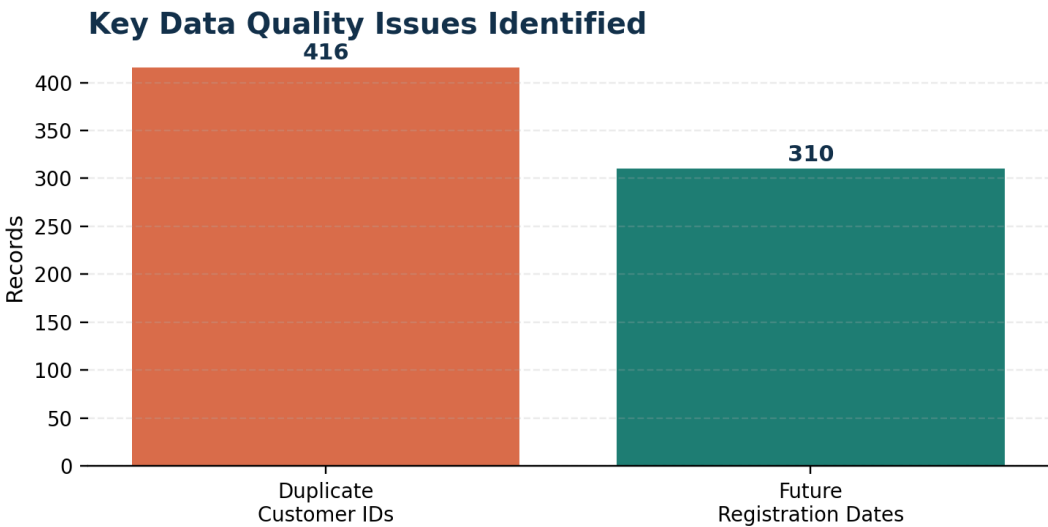
2. Dataset Summary

The dataset originally contained approximately 50,000 customer records collected over a 12-month period. Each row represents one customer record with demographic, purchase, engagement, and support-related information.

Category	Columns	Purpose
Customer Identity	Customer_ID, Name	Used for reporting and customer targeting only
Demographics	Age, Gender, Location	Provides customer profile context
Account Data	Registration_Date	Used to calculate customer tenure
Purchase Behavior	Last_Purchase_Date, Purchase_Frequency, Total_Spending, Average_Order_Value	Measures buying behavior and customer value
Engagement	Website_Visits, Email_Open_Rate	Measures digital engagement with the store
Service Interaction	Customer_Support_Tickets	Captures complaint or assistance activity

3. Data Quality Assessment

The dataset was generally usable, but several issues were identified and handled before modeling. These issues were important because duplicate records, invalid dates, and inconsistent categories can weaken model reliability if left untreated.



Issue	Finding	Decision
Duplicate Customer IDs	416 duplicate customer records were identified.	Sorted by Last_Purchase_Date and kept the most recent customer record.
Invalid Future Registration Dates	310 future registration dates were found.	Excluded from tenure calculations and treated as invalid.
Missing Email Open Rate	Missing values were present.	Created Email_Open_Rate_Missing flag and used imputation during modeling.
Inconsistent Locations	Location names had inconsistent formatting.	Standardized locations before analysis and modeling.
Non-predictive Fields	Customer_ID and Name do not describe behavior.	Excluded from model training but retained for reporting.

4. Feature-Level Analysis

4.1 Customer ID and Name

Customer_ID had no missing values and followed a valid identifier format. However, duplicate IDs were detected and resolved. Customer_ID and Name were retained for reporting, but excluded from training because they do not describe customer behavior.

4.2 Age

The Age column had no missing values. Customer ages ranged from 18 to 63 years, with an average age of approximately 31 years. Most customers belonged to the 25-36 age group. Purchase and engagement behavior were relatively similar across age groups.

Age Conclusion Age provides useful demographic context, but it is not a strong standalone churn indicator.
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4.3 Gender

The dataset was composed of approximately 69% female customers and 31% male customers. Behavior across gender groups was highly similar, with only minor differences in purchase frequency, email engagement, website visits, and support tickets.

Metric	Female	Male	Interpretation
Average Purchase Frequency	5.59	5.55	Very similar
Average Email Open Rate	32.92%	32.72%	Very similar
Average Website Visits	37.40	37.05	Very similar
Average Support Tickets	1.72	1.73	Very similar

4.4 Location

The Location column contained 19 unique locations with no missing values. Customers were concentrated mainly in major cities such as Karachi, Lahore, and Islamabad. Metrics across locations showed only minor differences, so Location was retained as a supporting feature rather than a primary churn driver.

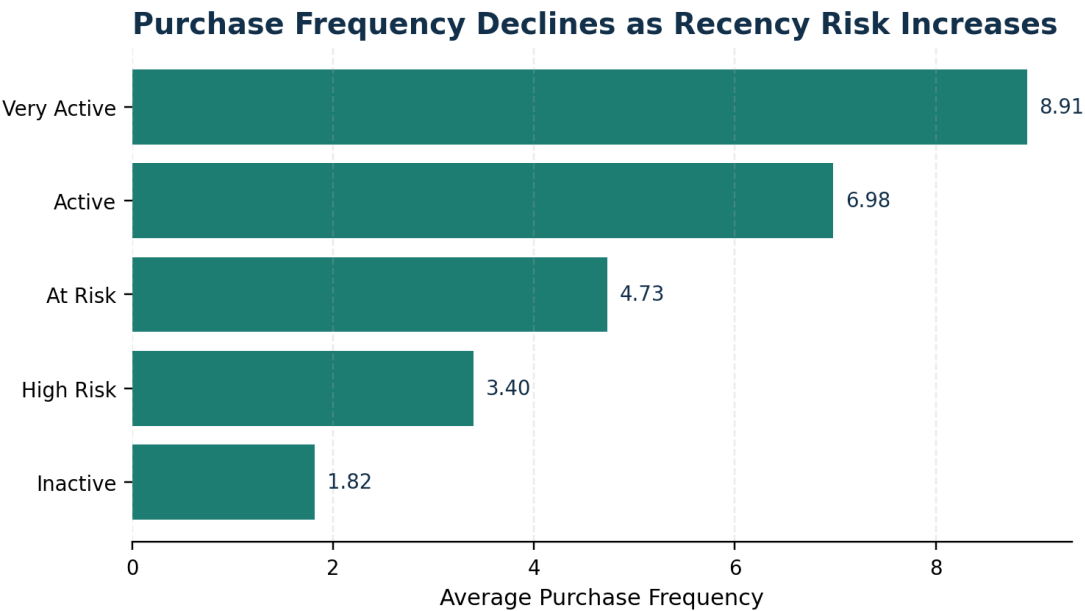
4.5 Registration Date and Customer Tenure

Registration_Date was used to create Customer_Tenure_Days. After excluding invalid future dates, customer tenure groups showed only small differences in purchase frequency, spending, website visits, support tickets, and email engagement.

Tenure Group	Purchase Frequency	Total Spending	Website Visits	Support Tickets	Email Open Rate
0-90 days	5.42	372.94	36.95	1.77	32.79%
90-180 days	5.60	383.44	37.32	1.74	32.71%
180-365 days	5.63	385.31	37.47	1.72	32.98%
365-730 days	5.59	381.28	37.35	1.72	32.90%

4.6 Last Purchase Date and Purchase Recency

Purchase recency was one of the most important signals in the dataset. Customers who purchased recently showed higher purchase frequency, higher total spending, stronger website activity, stronger email engagement, and fewer support tickets. Inactive customers showed the opposite behavior.



Recency Metrics Summary

Recency Group	Purchase Frequency	Total Spending	Website Visits	Email Open Rate	Support Tickets
Very Active	8.91	608.79	58.24	43.68%	1.24
Active	6.98	478.34	45.97	38.09%	1.38
At Risk	4.73	329.33	31.98	30.82%	1.78
High Risk	3.40	228.63	23.07	25.70%	2.09
Inactive	1.82	123.15	8.35	13.37%	3.13

Leakage Note

Because Days_Since_Last_Purchase was later used to define the churn target, it was removed from the final modeling features to prevent data leakage.

4.7 Purchase Frequency

Purchase_Frequency showed strong business value. Customers with higher purchase frequency were more engaged and valuable. High-frequency customers were not removed as outliers because they may represent loyal or VIP customers.

Purchase Frequency Conclusion

Purchase frequency is a key behavioral feature and was expected to be one of the strongest churn predictors.

4.8 Total Spending

Total_Spending was highly right-skewed, with a skewness value of approximately 2.98. High-spending customers were not removed because they represent important customer value. A log-transformed feature, Total_Spending_Log, was created for modeling.

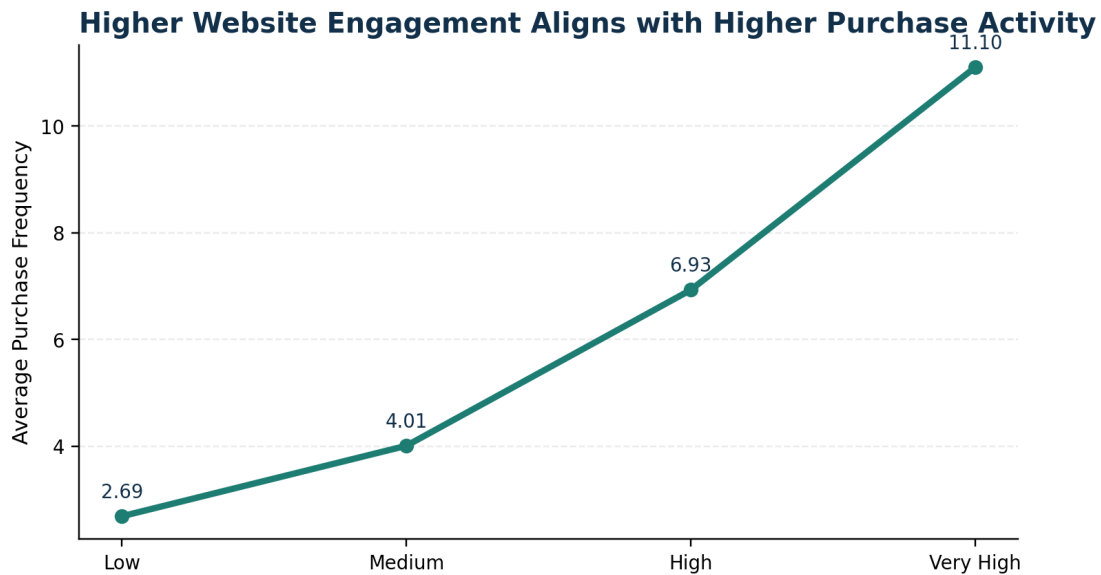
4.9 Average Order Value

Average_Order_Value had moderate right skewness, with a skewness value of approximately 1.8. A log-transformed feature was created. However, AOV segment analysis showed very small differences in churn-related behavior across Low, Medium, High, and Premium groups.

AOV Segment	Days Since Last Purchase	Support Tickets	Email Open Rate	Purchase Frequency
Low	73.52	1.71	34.35%	6.11
Medium	72.43	1.69	34.61%	6.04
High	73.44	1.69	34.44%	6.20
Premium	72.27	1.67	34.27%	6.15

4.10 Website Visits

Website_Visits was one of the strongest engagement indicators. Customers with low website activity had lower purchase frequency, lower spending, lower email engagement, more support tickets, and longer gaps since their last purchase.



Website Engagement	Purchase Frequency	Total Spending	Email Open Rate	Support Tickets	Days Since Last Purchase
Low	2.69	182.78	21.48%	2.34	126.48
Medium	4.01	275.37	28.60%	1.85	82.89
High	6.93	474.85	38.42%	1.48	48.32
Very High	11.10	756.95	49.51%	1.07	30.85

4.11 Email Open Rate

Email_Open_Rate reflects engagement with marketing campaigns. Missing values were handled carefully by creating an Email_Open_Rate_Missing indicator. Lower email engagement generally aligned with weaker customer activity and higher churn-related behavior.

4.12 Customer Support Tickets

Customer_Support_Tickets may indicate dissatisfaction or unresolved issues. Inactive customers had an average of 3.13 support tickets, compared with 1.24 for very active customers. This suggests support activity can be an early warning signal for churn risk.

5. Key Business Insights

- **Behavior matters most:** Purchase frequency, website visits, email engagement, and support interactions showed stronger churn patterns than demographics.
- **Low website activity is a warning sign:** Customers with low website visits showed much weaker purchase behavior and longer inactivity gaps.
- **Support ticket volume needs attention:** Customers with more support tickets were more likely to appear in inactive or high-risk groups.
- **Demographics are supporting variables:** Age, gender, and location provide context but are not strong standalone churn predictors.
- **Data leakage was identified and avoided:** Days_Since_Last_Purchase was excluded from model training because it was used to define churn.

6. Features Recommended for Modeling

Recommended Feature	Reason for Inclusion
Age	Provides demographic context
Gender	Supporting demographic feature
Location	Supporting geographic context
Customer_Tenure_Days	Measures customer relationship length
Purchase_Frequency	Strong behavioral churn signal
Average_Order_Value	Measures customer order value
Average_Order_Value_Log	Reduces skewness in AOV
Website_Visits	Strong engagement signal
Email_Open_Rate	Marketing engagement signal
Email_Open_Rate_Missing	Captures missing engagement behavior
Customer_Support_Tickets	Potential dissatisfaction signal
Total_Spending_Log	Captures customer value while reducing skewness

Excluded Columns

Excluded Column Type	Examples	Reason
Identifiers	Customer_ID, Name	Used for reporting, not prediction
Raw Dates	Registration_Date, Last_Purchase_Date	Converted into engineered features
EDA Grouping Columns	Age_Group, AOV_Segment, Website_Engagement	Created for analysis only, not model training
Leakage Feature	Days_Since_Last_Purchase	Used to define churn target, so excluded from training

7. EDA Conclusion

The exploratory data analysis showed that customer churn risk is primarily associated with behavioral and engagement patterns. Customers who purchase less frequently, visit the website less often, open fewer emails, and submit more support tickets are more likely to show churn-related behavior.

The analysis also identified and resolved important data quality issues, including duplicate customer records, invalid future registration dates, missing engagement values, and inconsistent categorical fields. These cleaning and feature engineering steps improved dataset reliability and prepared it for machine learning.

Overall, the dataset contains meaningful behavioral patterns that support the business objective of predicting churn and enabling targeted retention campaigns.

8. Next Steps

- Create a churn target variable using the agreed business definition.
- Prepare the final model-ready feature set.
- Train and compare multiple classification models.
- Evaluate models using Recall, F1 Score, and ROC AUC.
- Generate customer risk segments for marketing action.
- Deploy the solution through a Streamlit dashboard for business use.

Final Business Value

The EDA confirms that Sarah Ahmed’s marketing team can move from broad discounting to targeted retention by focusing on customers with weak purchase behavior, low digital engagement, and high support interaction.